

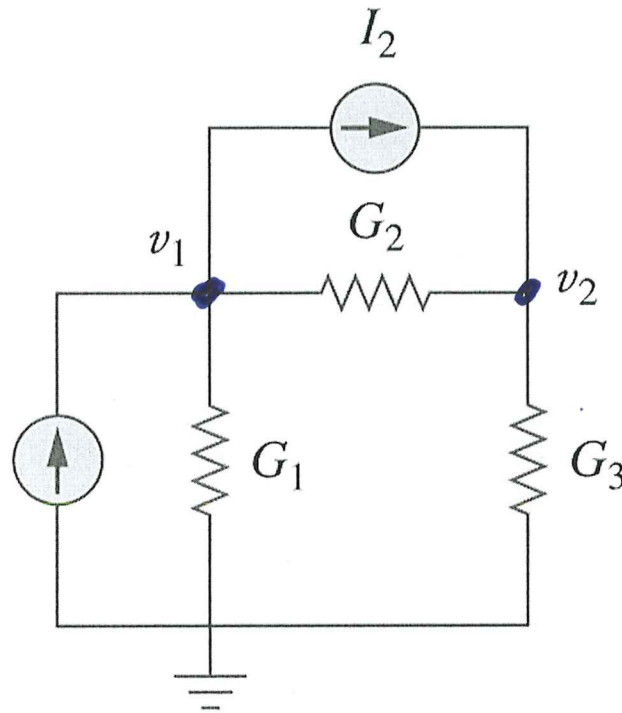
L#11: Nodal and mesh analysis by inspection

Nodal analysis by inspection

2017-08-10

Diagonal

$$C_{ij} = C_{ji} \quad i \neq j$$



$n \times n$
0

1. Format; 2. Conductance matrix; 3. Direction of independent source; 4. Applicability.

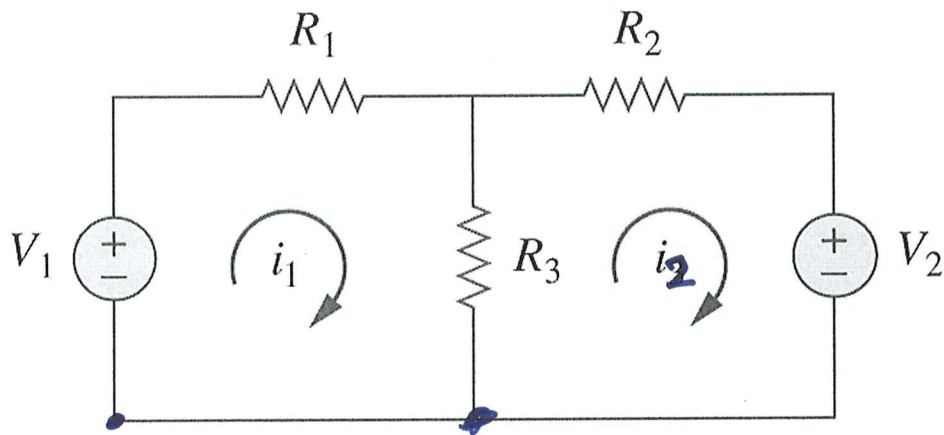
$$\begin{pmatrix} G_1 + G_2 & -G_2 \\ -G_2 & G_2 + G_3 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} I_1 - I_2 \\ I_2 \end{pmatrix}$$

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conductance matrix
nodal voltage
vector

sum of independent
current source
entering a
node

Mesh analysis by inspection

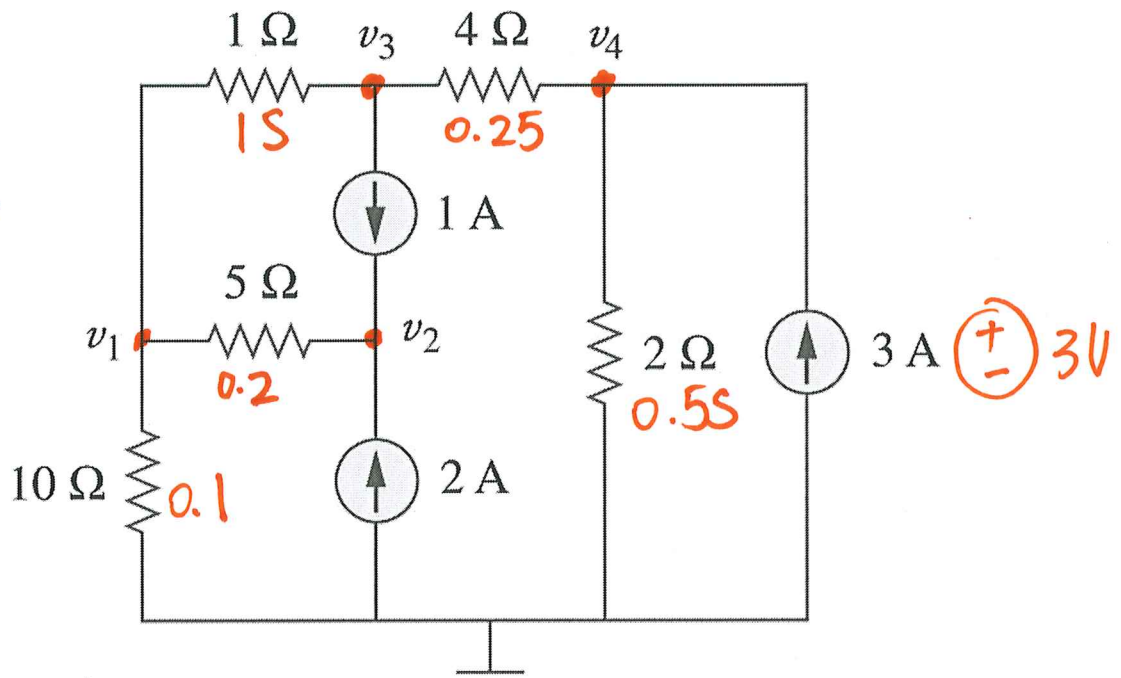


1. Format; 2. Resistance matrix; 3. Direction of independent source; 4. Applicability.

$$\underbrace{\begin{pmatrix} R_1 + R_3 & -R_3 \\ -R_3 & R_2 + R_3 \end{pmatrix}}_{\text{Resistance matrix}} \underbrace{\begin{pmatrix} i_1 \\ i_2 \end{pmatrix}}_{\text{vector for mesh current}} = \begin{pmatrix} -V_1 \\ -V_2 \end{pmatrix} \leftarrow \begin{array}{l} \text{sum of} \\ \text{independent} \\ \text{voltage increase} \\ \text{of a mesh} \end{array}$$

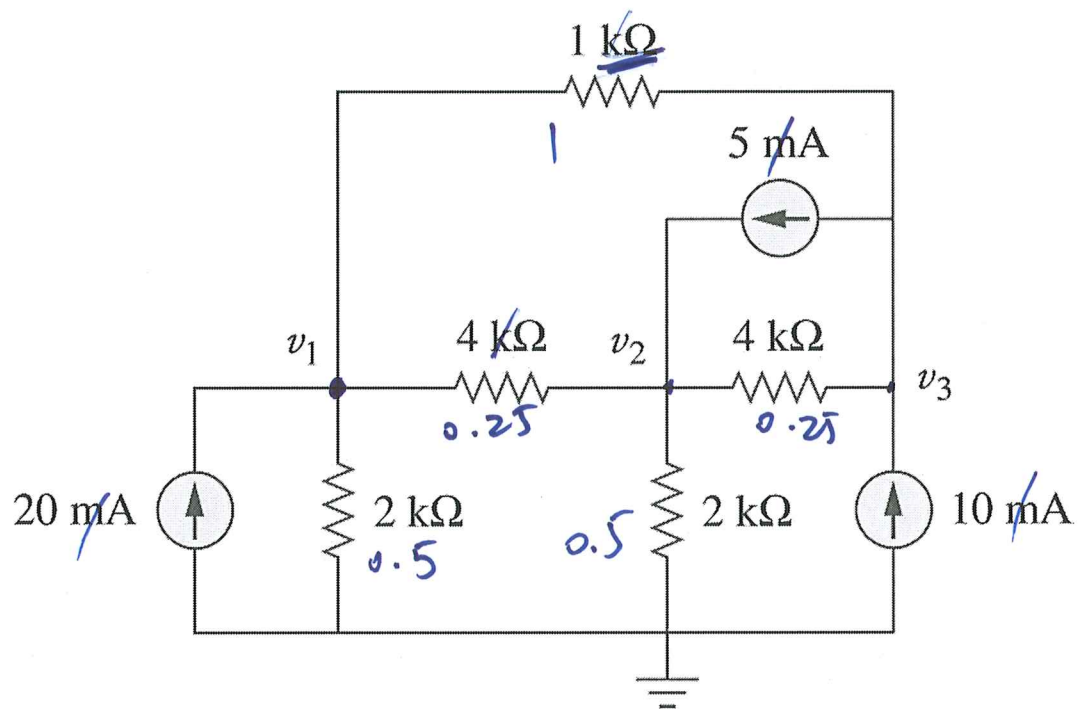
Example: Nodal analysis by inspection (1)

4x4



$$\begin{pmatrix} 1+0.2+0.1 & -0.2 & -1 & 0 \\ -0.2 & 0.2 & 0 & 0 \\ -1 & 0 & 1.25 & -0.25 \\ 0 & 0 & -0.25 & 0.75 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 1+2 \\ -1 \\ 3 \end{pmatrix}$$

Example: Nodal analysis by inspection (2)

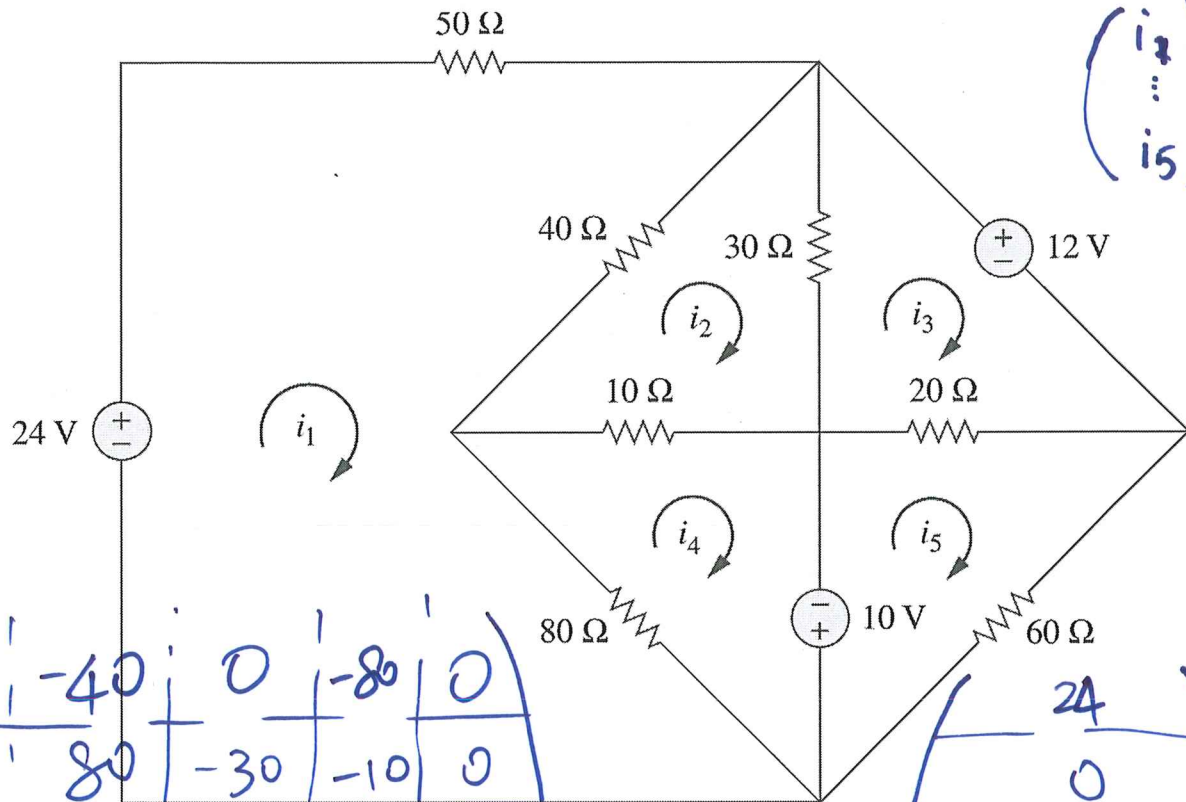


$$\begin{pmatrix} 1.75 & -0.25 & -1 \\ -0.25 & 1 & -0.25 \\ -1 & -0.25 & 1.25 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} = \begin{pmatrix} -20 \\ 5 \\ 10 \end{pmatrix}$$

Example: Mesh analysis by inspection

(A)

$\begin{pmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{pmatrix}$



$$\begin{pmatrix} 170 & -40 & 0 & -80 & 0 \\ -40 & 80 & -30 & -10 & 0 \\ 0 & -30 & 50 & 0 & -20 \\ -80 & -10 & 0 & 90 & 0 \\ 0 & 0 & -20 & 0 & 80 \end{pmatrix}$$

$\times (A) =$

$$\begin{pmatrix} 24 \\ 0 \\ -12 \\ 10 \\ -10 \end{pmatrix}$$